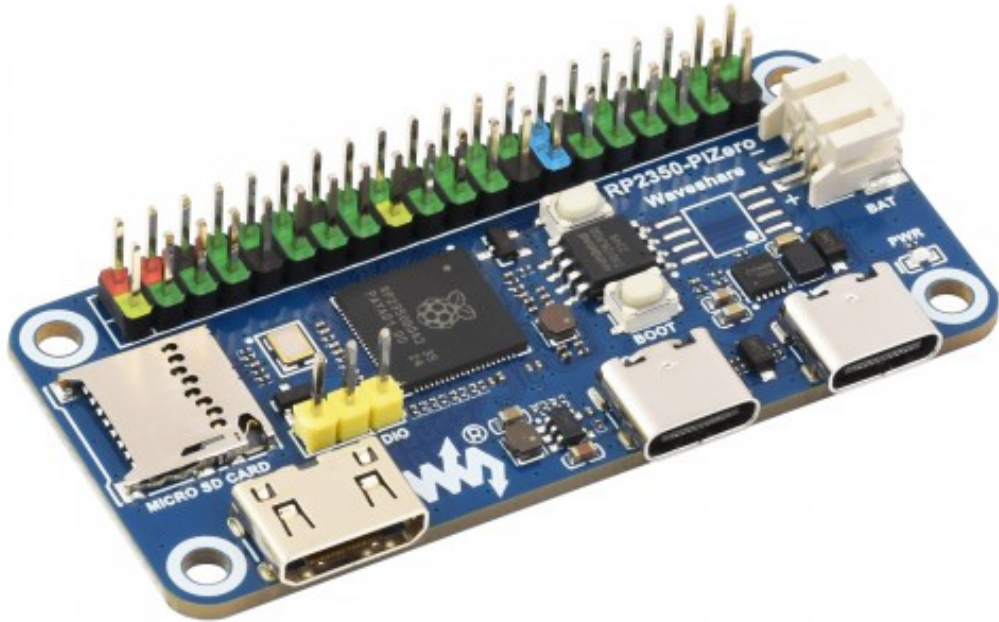




[Raspberry Pi Pico 2](#)

Waveshare RP2350-PiZero Development Board, Based On The Raspberry Pi RP2350 Dual-core & Dual-architecture Microcontroller, 520KB SRAM And 16MB Onboard Flash Memory

SKU 31836
Part No. RP2350-PiZero
Brand Waveshare

CA\$14.²¹

1

ADD TO CART

Price

Quantity

CA\$14.21

1

CA\$13.64

 Chat ²

CA\$13.36

3

CA\$13.25

4+

Related Products:

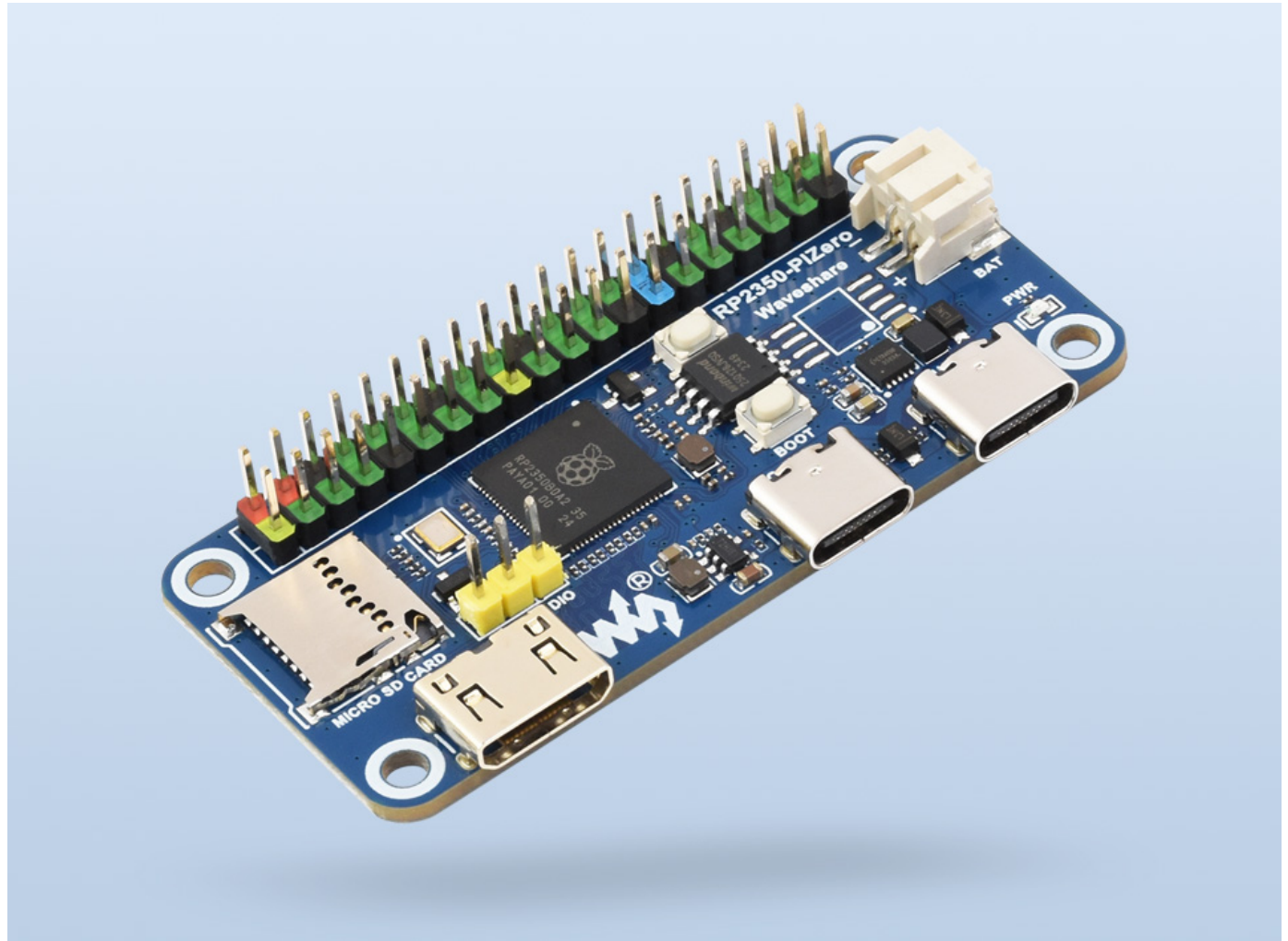


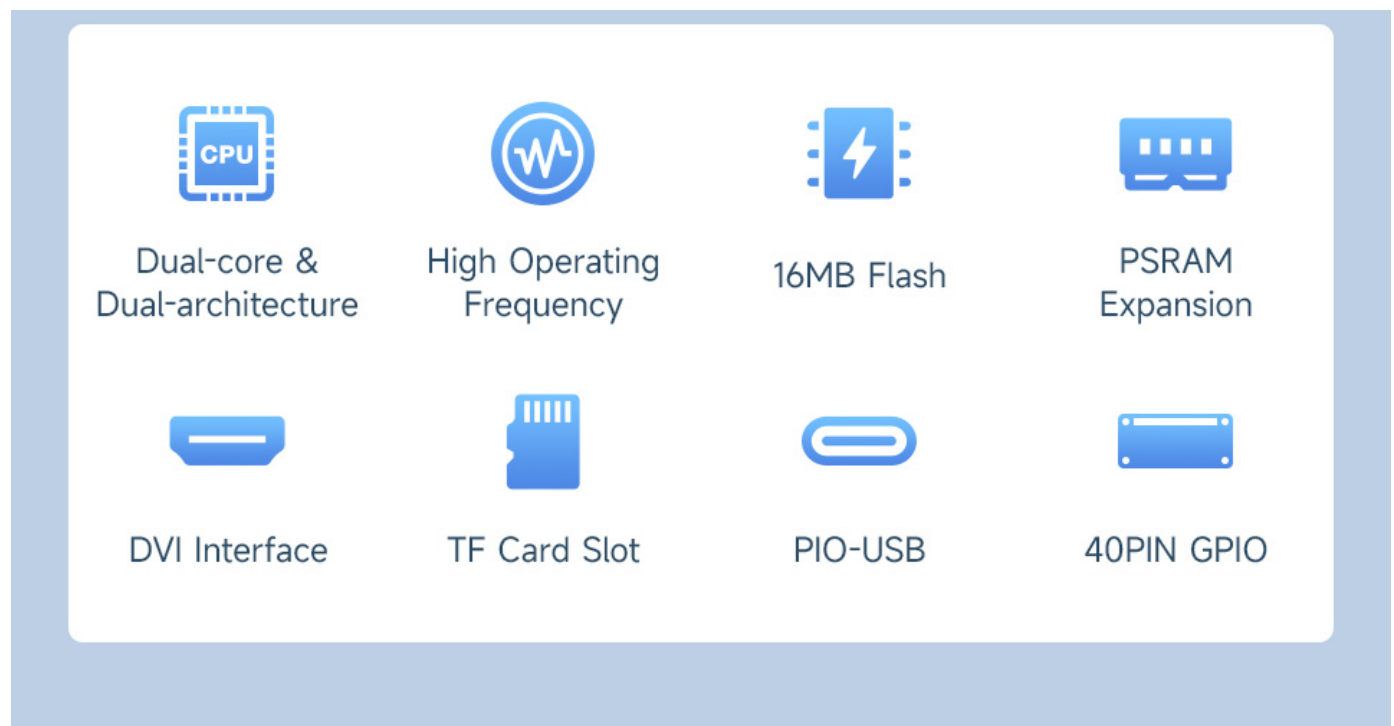
Description	Package Content	Wiki
Waveshare RP2350-PiZero Development Board, Based On The Raspberry Pi RP2350 Dual-Core & Dual-Architecture Microcontroller, 520KB SRAM And 16MB Onboard Flash Memory, Compatible With Raspberry Pi 40PIN GPIO Header		
Chat		

RP2350-PiZero Development Board

Based On Raspberry Pi RP2350

Onboard DVI Interface, TF Card Slot And PIO-USB Port





Board Specifications

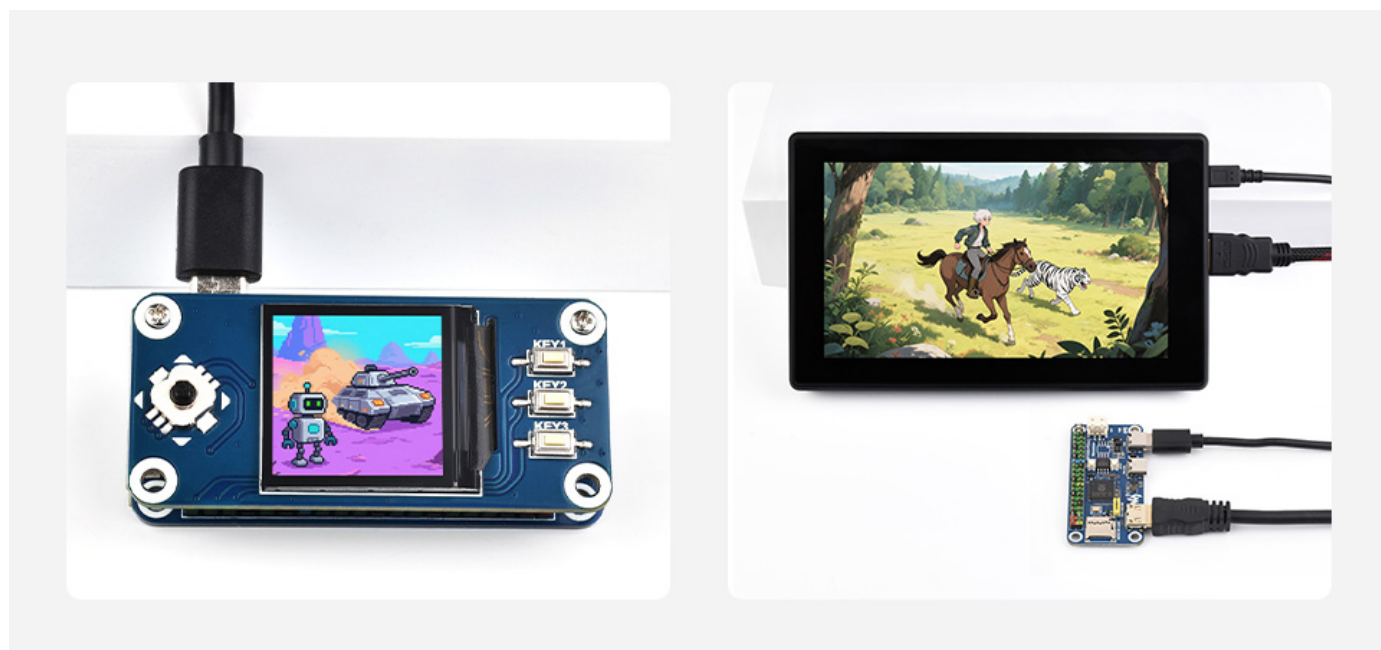
RP2350-PiZero is a high-performance and cost-effective microcontroller board designed by Waveshare, onboard DVI interface, TF card slot and PIO-USB port, compatible with Raspberry Pi 40PIN GPIO header, with reserved solder pads for PSRAM chip expansion, easy to develop and integrate into products.

- RP2350B microcontroller chip designed by Raspberry Pi in the United Kingdom
- Adopts unique dual-core and dual-architecture design: dual ARM Cortex-M33 or dual Hazard3 RISC-V processors support, flexible clock running up to 150 MHz
- 520KB of SRAM, and 16MB of onboard Flash memory, with reserved solder pads for PSRAM chip expansion
- Type-C connector for easier use
- Onboard DVI interface can drive most HDMI screens (DVI compatibility required)
- Supports using as a USB host or device via onboard PIO-USB port
- Onboard TF card slot for reading and writing TF card
- Onboard Lithium battery recharge/discharge header, suitable for mobile scenarios
- USB 1.1 with device and host support
- Low-power sleep and dormant modes
- Drag-and-drop programming using mass storage over USB

- Adapting 5 × multi-function GPIO pins
- 2 × SPI, 2 × I2C, 2 × UART, 16 × controllable PWM channels
- Accurate clock and timer on-chip
- Temperature sensor
- Accelerated floating-point libraries on-chip
- 12 × Programmable I/O (PIO) state machines for custom peripheral support

Integrates Multiple Interfaces

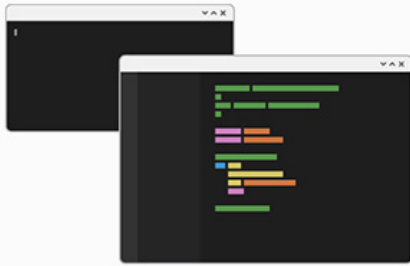
Onboard DVI Interface, PIO-USB Port And TF Card Slot For More Possibilities



for reference only, the [1.3inch LCD HAT](#) and [7inch HDMI LCD \(H\)](#) are Not included

Support C/C++, MicroPython, Arduino IDE

Comprehensive SDK, Dev Resources, Tutorials To Help You Easily Get Started



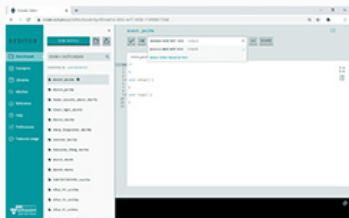
Pico C/C++ SDK

The Raspberry Pi official C SDK can be used from the command line, or from popular integrated development environments like VS Code and Eclipse.



MicroPython

Micropython is a full implementation of the Python 3 programming language that runs directly on embedded hardware like Raspberry Pi Pico.



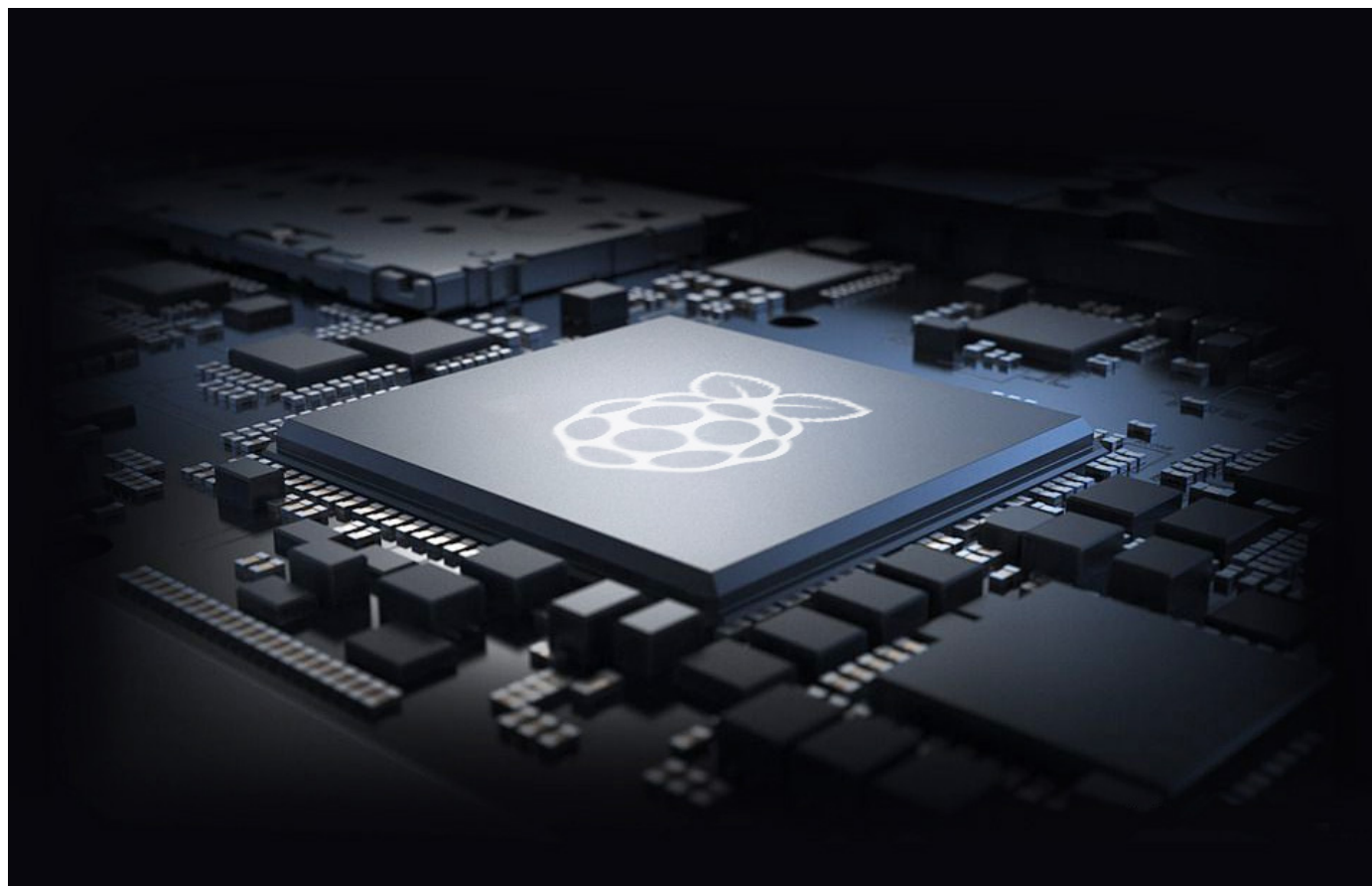
Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started.

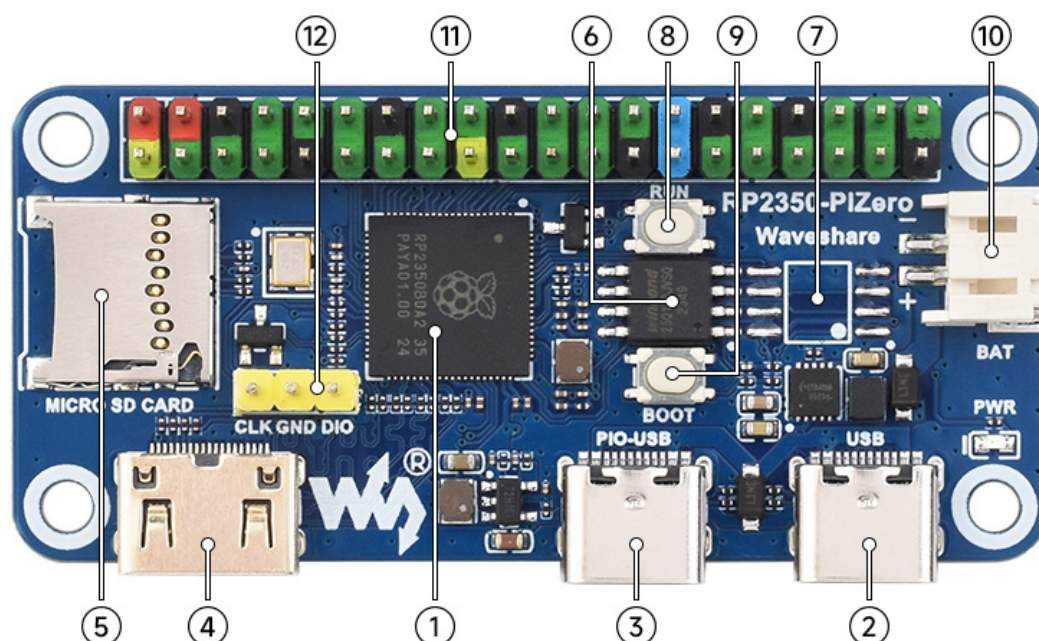
* the DVI and PIO-USB functions are only supported in C environment and cannot be used at the same time

Dual-Core, Dual-Architecture

The Unique Dual-Core, Dual-Architecture Capability Of RP2350B Allows Users To Choose Between A Pair Of Industry-Standard Arm Cortex-M33 Cores, And A Pair Of Open-Hardware Hazard3 Cores, Giving Developers A Chance To Experiment With The RISC-V Architecture In A Stable, Well-Supported Environment



What's On Board



1. RP2350B

Dual-core and dual-architecture design, up to 150 MHz operating frequency

2. USB Type-C connector

for programming, USB 1.1 with device and host support

3. PIO-USB port

for using as a USB host or device via PIO

4. DVI interface (Mini HDMI connector)

for driving DVI screen with HDMI cable

5. TF card slot

supports SPI and SDIO interface communication

6. 16MB NOR-Flash

7. PSRAM chip solder pads

Compatible with mainstream PSRAM chips for user expansion (soldering required)

8. RUN button

reset button

9. BOOT button

Press it when resetting to enter download mode

10. Lithium battery header

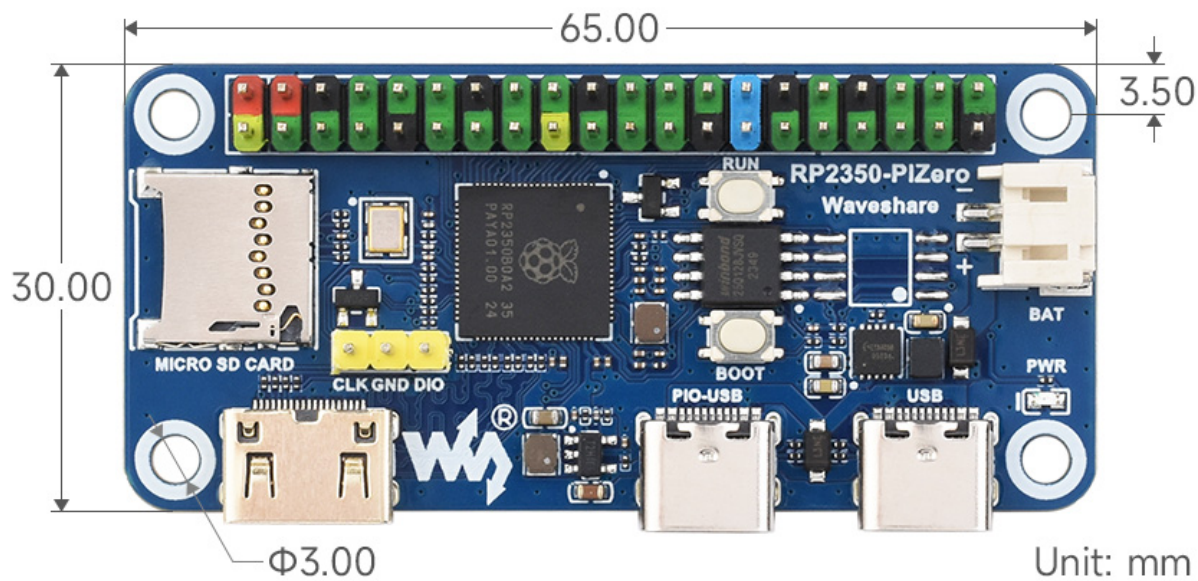
PH2.0 header, for connecting 3.7V Lithium battery, allows recharging/discharging the battery

11. 40PIN GPIO header

Compatible with some Raspberry Pi HATs

12. Debug interface

Outline Dimensions



Resources & Services



WAVESHARE
ORIGINAL



PROFESSIONAL
TECH SUPPORT



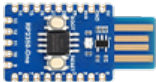
RICH RESOURCES
MANUAL & DEMO

Wiki: www.waveshare.com/wiki/RP2350-PiZero

Pico Product Selection

PRODUCT	VOLTAGE REGULATOR CHIP	USB	FLASH	LITHIUM BATTERY HEADER	FEATURES
RP2350 CHIP					
 RASPBERRY PI PICO 2	DC-DC, 800mA	Micro USB	4MB	-	-
 RASPBERRY PI PICO 2 W	DC-DC, 800mA	Micro USB	4MB	-	with WIFI and Bluetooth

Chat

 RASPERRY PI PICO 2 WH	DC-DC, 800mA	Micro USB	4MB	-	with WIFI and Bluetooth
 CAN VER.	DC-DC, 2000mA	Type-C	4MB	-	CAN interface
 TYPE-A VER.	LDO, 500mA	Type-A	4MB	-	-
 TYPE-C FEMALE VER.	LDO, 800mA	Type-C	4MB	-	-
 TYPE-C MALE VER.	LDO, 800mA	Type-C	2MB	-	-
 PIO-USB FEMALE VER.	LDO, 500mA	Type-C	2MB	-	with PIO-USB Type-C female port
 PIO-USB MALE VER.	LDO, 500mA	Type-C	2MB	-	with PIO-USB Type-C male port
 TINY VER.	LDO, 800mA	Separate Type-C port adapter	4MB	-	Separate adapter board design, adapting USB Type-C port via FPC
 RGB MATRIX VER.	LDO, 800mA	Type-C	16MB	-	Onboard 8 x 8 RGB LED Matrix
 ETH PORT VER.	LDO, 800mA	Type-C	4MB	-	with Ethernet port
 POE VER.	LDO, 800mA	Type-C	16MB	-	with Ethernet port; Supports PoE expansion via a PoE Module
 USB VER.	LDO, 500mA	Type-A, Type-C	2MB	-	with PIO-USB Type-A and USB Type-C port



CORE BOARD

LDO, 800mA

Separate Type-C
port adapter

16MB

-

-



WIRELESS PLUS VER.

LDO, 800mA

Type-C

16MB

-

Supports Wi-Fi and Bluetooth

1.43INCH ROUND
TOUCH AMOLED VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.43inch AMOLED Touch LCD

1.64INCH TOUCH
AMOLED VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.64inch AMOLED Touch LCD

1.75INCH ROUND
TOUCH AMOLED VER.DC-DC, 2000mA
LDO,300mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.75inch AMOLED Touch LCD

1.8INCH TOUCH
AMOLED VER.DC-DC, 2000mA
LDO, 300mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.8inch AMOLED Touch LCD

2.41INCH TOUCH
AMOLED VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2.41inch Touch AMOLED



0.85INCH LCD VER.

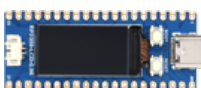
DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

0.85inch IPS LCD



0.96INCH LCD VER.

DC-DC, 2000mA

Type-C

4MB

Support(allows recharging the
battery and powering the board at
the same time)

0.96inch IPS LCD

Chat



1.28inch ROUND LCD
VER.

LDO, 300mA

Type-C

4MB

Support(allows recharging the
battery and powering the board at
the same time)

1.28inch IPS LCD



1.28inch ROUND
TOUCH LCD VER.

LDO, 300mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.28inch IPS touch LCD



1.46inch ROUND
TOUCH VER.

DC-DC, 2000mA

Type-C

16MB

-

1.46inch touch LCD



1.47inch LCD VER.

LDO, 300mA

Type-C

16MB

-

1.47inch IPS LCD



1.47inch LCD USB
VER.

LDO, 300mA

Type-A

16MB

-

1.47inch IPS LCD



1.54inch LCD VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.54inch IPS LCD



1.54inch TOUCH LCD
VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.54inch IPS Touch LCD



1.69inch TOUCH LCD
VER.

LDO, 500mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.69inch Touch LCD

Chat



1.85INCH ROUND
TOUCH LCD VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.85inch Touch LCD



2INCH TOUCH LCD
VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2inch Touch LCD



2.1INCH ROUND
TOUCH LCD VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2.1inch Touch LCD



2.1INCH ROUND
CURVED TOUCH LCD
VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2.1inch 2.5D Curved Touch LCD



2.8INCH TOUCH LCD
VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2.8inch IPS Touch LCD



2.8INCH ROUND
TOUCH LCD VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

2.8inch IPS Touch LCD, RGB
interface



3.49INCH TOUCH LCD
VER.

DC-DC, 2000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

3.49inch IPS Touch LCD



3.5INCH TOUCH LCD
VER.

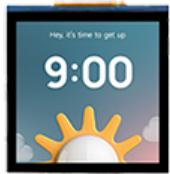
DC-DC, 2000mA
LDO, 300mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

3.5inch Touch LCD



4INCH LCD VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

4inch IPS LCD, RGB interface



4INCH TOUCH LCD
VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

4inch IPS Touch LCD, RGB
interface



4.3INCH TOUCH LCD
VER.

LDO, 800mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

4.3inch IPS Touch LCD, RGB
interface



7INCH TOUCH LCD
VER.

DC-DC, 3000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

7inch IPS Touch LCD, RGB
interface



7INCH LCD VER.

DC-DC, 3000mA

Type-C

16MB

Support(allows recharging the
battery and powering the board at
the same time)

7inch IPS LCD, RGB interface



1.54INCH E-PAPER
VER.

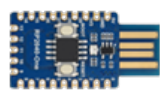
DC-DC, 2000mA

Type-C

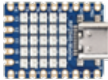
16MB

Support(allows recharging the
battery and powering the board at
the same time)

1.54inch e-Paper display

 1.54INCH TOUCH E-PAPER VER.	DC-DC, 2000mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	1.54inch touch e-Paper display
 PLUS VER.	DC-DC, 2000mA	Type-C	4MB/16MB	Support(allows recharging the battery and powering the board at the same time)	-
 DVI PORT VER.	DC-DC, 2000mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	Onboard DVI interface, TF card slot and PIO-USB port
RP2040 CHIP					
 RASPERRY PI PICO W	DC-DC, 3000mA	Micro USB	2MB	-	with WIFI
 RASPERRY PI PICO WH	DC-DC, 3000mA	Micro USB	2MB	-	with WIFI
 RASPERRY PI PICO	DC-DC, 800mA	Micro USB	2MB	-	-
 RASPERRY PI PICO H	DC-DC, 800mA	Micro USB	2MB	-	-
 TYPE-A VER.	LDO, 500mA	Type-A	4MB	-	-
 MINI VER.	LDO, 800mA	Type-C	2MB	-	-
 TINY VER.	LDO, 500mA	Separate Type-C port adapter	2MB	-	Separate adapter board design, adapting USB Type-C port via FPC

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 RGB MATRIX VER.	LDO, 500mA	Type-C	2MB	-	Onboard 5×5 RGB LED matrix
 BLE VER.	LDO, 300mA	Separate Type-C port adapter	2MB	-	Supports Bluetooth 5.1, separate adapter board design, adapting USB Type-C port via FPC
 ETH PORT VER.	LDO, 500mA	Type-C	4MB	-	with Ethernet port
 0.99INCH ROUND LCD WITH CASE VER.	LDO, 500mA	Type-C	2MB	-	0.99inch IPS LCD, CNC metal case
 1.28INCH ROUND LCD VER.	LDO, 300mA	Type-C	2MB	Support(allows recharging the battery and powering the board at the same time)	1.28inch IPS LCD
 1.28INCH ROUND LCD WITH CASE VER.	LDO, 300mA	Type-C	2MB	Support(allows recharging the battery and powering the board at the same time)	1.28inch IPS LCD, CNC metal case
 1.28INCH ROUND TOUCH LCD VER.	LDO, 500mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	1.28inch Touch LCD
 1.28INCH ROUND TOUCH LCD WITH CASE VER.	LDO, 500mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	1.28inch Touch LCD, CNC metal case
 1.69INCH TOUCH LCD VER.	LDO, 500mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	1.69inch Touch LCD

 1.85INCH TOUCH LCD VER.	DC-DC, 2000mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	1.85inch Touch LCD
 0.96 LCD VER.	DC-DC, 1800mA	Type-C	2MB	Support(allows recharging the battery and powering the board at the same time)	0.96inch IPS LCD
 PLUS VER.	DC-DC, 1800mA	Type-C	4MB/16MB	Support(allows recharging the battery and powering the board at the same time)	-
 DVI PORT VER.	LDO, 300mA	Type-C	16MB	Support(allows recharging the battery and powering the board at the same time)	Onboard DVI interface, TF card slot and PIO-USB port
 CAM & LCD VER.	-	Type-C	16MB	-	Onboard grayscale camera and 1.14inch IPS LCD display